

7 6. Power trouble

When you're building something that eats so much power, you're bound to meet this roadblock. As it happens, our Ultimate PC runs on two PSUs with a combined wattage of a whopping 2300 Watts, the likes of which we've never witnessed before and will probably never need. But with so much electricity needed by the PC, we had to make sure we ran it on a separate circuit breaker altogether. One of our spike strips sacrificed itself when we connected both the PSU power cables to it and so we had to use two separate spike strips for the job. Now that's what happens when you play with more power than you can possibly handle.

8 A case of fat GPUs and shortage of PCIe ports

Call it first world problems or something else but when you have four full sized, dual slot GPUs, they take up almost all the space you can find on the motherboard and this was also the case with our Rampage IV Black Edition. With each card occupying two slots, we filled up all the four PCIe x16 slots with the GPUs and didn't have space left for anything else. This is precisely why we thought to make two builds with the same PC. One with the OCZ Z-Drive R4 PCIe SSD installed next to three GTX 780 Ti's in SLI and the other build with four GTX 780 Ti's in SLI excluding the OCZ drive from the build and using the SanDisk Extreme II SSDs for storage. Remember to measure everything before purchasing your components.

9 Unavailability of games

This can't be true right? Yes it can! We're talking about the unavailability of games that make full use of the hardware installed in the Ultimate PC. Though games are becoming increasingly realistic, most of them that are released these days are console ports. The problem with this is that even though these games might be good, they aren't developed with the point of view of making full use of a PC's resources. Many games still make use of only a single CPU core as opposed to making use of all the cores available. Something that the developers should improve upon.

10 Special drivers needed:

A special SSD calls for special drivers and the king of all drives - the OCZ Z-Drive R4 needed that too. It's not everyday that you use an enterprise-grade PCI Express SSD and in order for it to work we had to make sure that we installed separate drivers for this drive during the Windows 7 installation process itself. Since the OCZ drive is primarily built for servers, once you power on the system, the drive has its own separate POST process which increases the boot-up time of the PC.

11 Cable Management

Cables, Cables, Cables. A legion of cables. This is something you'll come across when you have to connect a bountiful of components and that too while using two PSUs. We used fully modular PSUs but that doesn't mean we had less cables to take care of. In fact, when we come to think of it, the maximum time we took during the entire build process was for cable management. Cable Management in a rig of this scale is very vital as you're not only worried about airflow but also about the overall looks. After spending hours in tying and tucking cables out of the way, we're rewarded with a PC that looks professionally assembled and spiffy.

12 Thermal problems

Since the Ultimate PC was chock full of awesome hardware, we had to make sure we prepared for several thermal issues that might arise later on. The installation of two PSUs meant that we had to find a way to provide airflow to both of them. The only solution to this was to face both the PSU fan sides opposite to each other and facing towards the sides of the case. This allowed for fresh air to be blown inside the PSU directly from the sides of the Corsair 900D. The next thermal mountain we had to climb was for the graphics cards because a quad SLI setup is like a heat sandwich in a PC. We had to install two additional 120mm fans on the drive cages to push cool air into the GPUs as we didn't want them to run hot while starving for air..

13 Miscellaneous

You need to always make sure that you reset the BIOS before you start a PC like this especially since the OC Panel that gets connected to the ASUS Rampage IV BE motherboard may have an OC profile activated, a lesson we learned the hard way (there was smoke). It's also vital to test all the components before hand, especially the PSUs. You don't want to lose a component worth upwards of 50k simply because you forgot to test some simple cable. Double check the connectors that you're using too and make sure that they're connected securely and in the correct slots. This will prevent you from accidentally blowing something up.

Complete Build Specifications:

Component	Brand and Model	Price (in ₹)
Processor	Intel Core i7 4960x Extreme Edition	78,000
Motherboard	ASUS Rampage IV Black Edition	35,200
RAM	Corsair Dominator Platinum 32GB 2133Mhz (CMD32GX3M4A2133C9)	38,999
Graphics Card	3x NVIDIA GeForce GTX 780Ti (Tri SLI)	1,80,000
Case	Corsair Obsidian 900D	28,999
Power Supply	Antec HCP Platinum 1300 Watts + 1000 Watts	47,000
CPU Cooler	NZXT Kraken X60	8,273
SSD	OCZ Z-Drive R4 CM88 1.8TB PCIe	9,90,000
SSD	SanDisk Extreme II 240GB x4	1,12,200
Monitor	Benq BL3200PT	75,000
Keyboard	Corsair Vengeance K70	9,999
Mouse	Razer Ouroboros	10,790
Headphones	Sennheiser G4ME Zero	25,000
Fans	3x120mm Cooler Master Sickle Flow X Red LED/ 2x120mm Corsair SP120L / 2x140mm Corsair	5,700
Lights	NZXT Sleeved LED Kit 200cm (Red) + Sunbeamtech Focus LEDs	1,148
Grand Total:		16,46,308

Alternate components used:

Component	Brand and Model	Price (in ₹)
RAM	Kingston Hyper X Predator 1866 MHz (KHX18C10T2K2/16X) x 2 kits	16,000
Graphics Card	1xGalaxy GTX 780 Ti HOF (added to existing 3x NVIDIA GTX 780 Ti)	60,000
PSU	Corsair AX1200i	26,000
Monitor	ASUS PB278	41,200

Conclusion:

Remember what we said about overkill being an understatement? Well, this PC's performance is simply staggering and completely unnecessary at the same time. Its performance figures are through the roof, but there's little real world application for it. It's testament to the system's power that we cracked the 3DMark's top 100 list simply by existing. What will happen once we actually overclock the system is something we don't yet know, but we're really excited to find out. Keep an eye on our website and facebook page to track our progress. For now, you can go to dgit.in/UltimatePC for more high-res pics of this monster in action. 